NPTEL ONLINE CERTIFICATION COURSES

Course Name: Software Project Management

Faculty Name: RAJIB MALL

Department : Computer Science and Engineering

TOPIC: Introduction

About Myself

- RAJIB MALL
- B.E. , M.E., Ph.D from Indian Institute of Science, Bangalore
- Worked with Motorola (India)
- Shifted to IIT, Kharagpur in 1994
  - Currently Professor at CSE Dept.

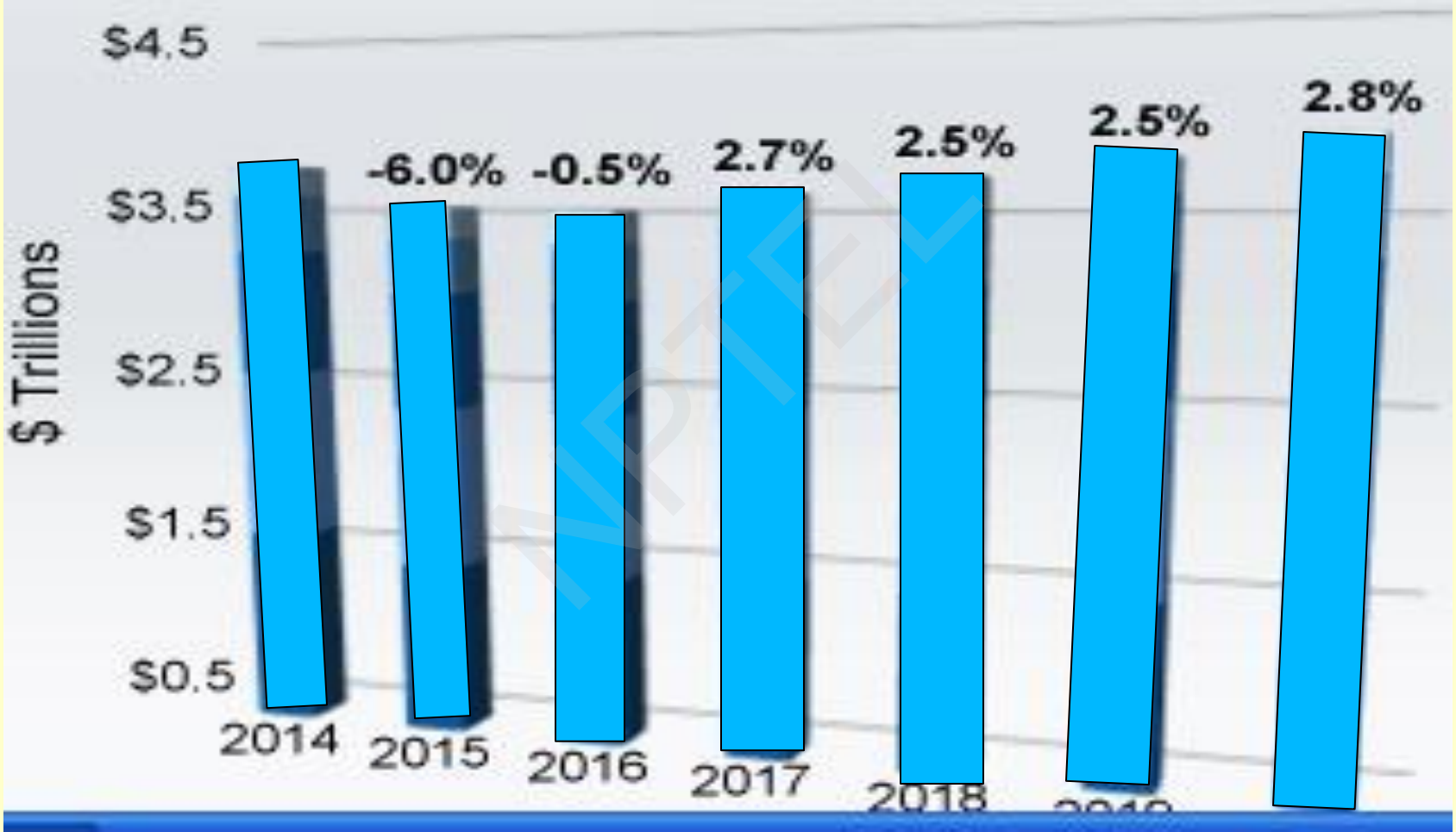




CONCEPTS COVERED

- What is software project management?
- Is software project management any different from traditional project management?
- Jobs, projects, and exploration
- What makes software project planning difficult?
- Project scope

IT Spending Worldwide (US $): Gartner



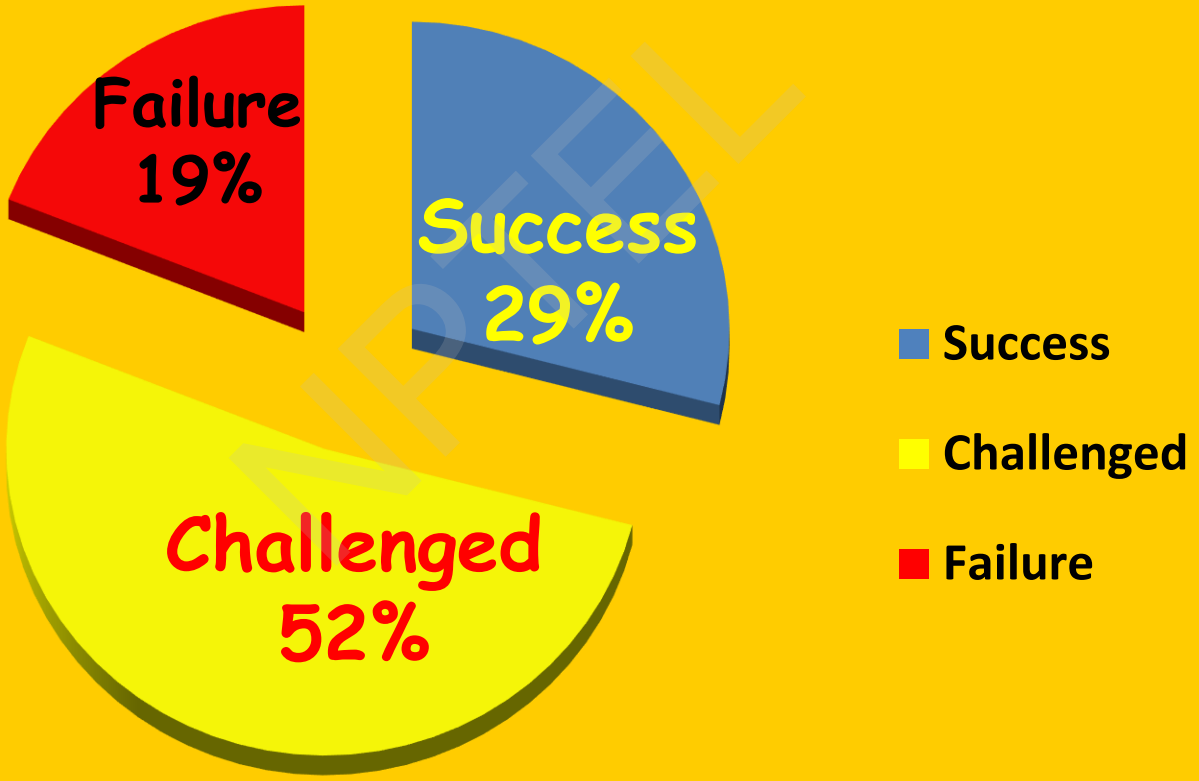
Software Crisis…

- We have a software crisis!
- What are the symptoms of the software crisis?
  - Fail to meet user requirements.
  - Expensive.
  - Difficult to alter, debug, and enhance.
  - Often delivered late.

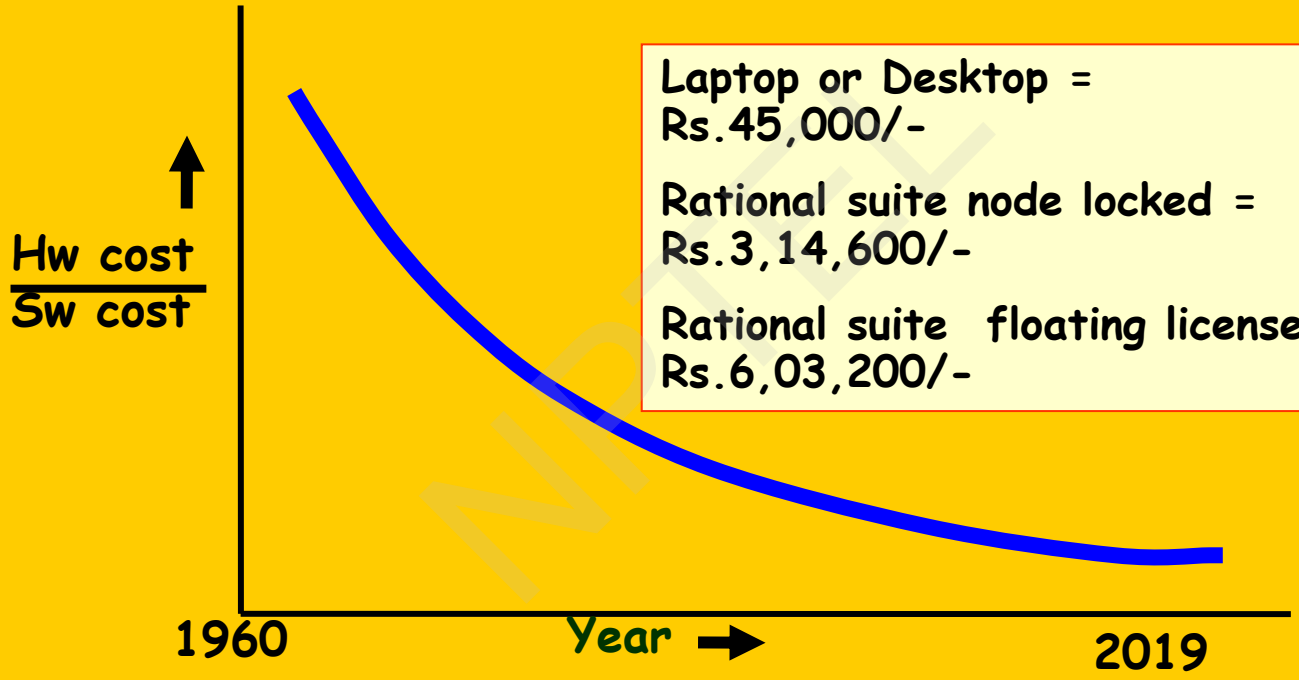
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Standish Group CHAOS Report

All projects



Software Crisis (cont.)



Laptop or Desktop = Rs.45,000/-

Rational suite node locked = Rs.3,14,600/-

Rational suite floating license= Rs.6,03,200/-

Relative Cost of Hardware and Software

Then why not have entirely hardware systems?...

- Virtue of software:
  - Relatively easy to change…
  - Consumes no space, weight, or power…
- Otherwise all might as well be hardware.

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Software Characteristics

- Two important characteristics:
  - The more is the complexity of software, the harder it is to change --- why?
  - The more the changes made to a program, the greater becomes its complexity.

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Which Factors are the Key Contributors to the Software Crisis?

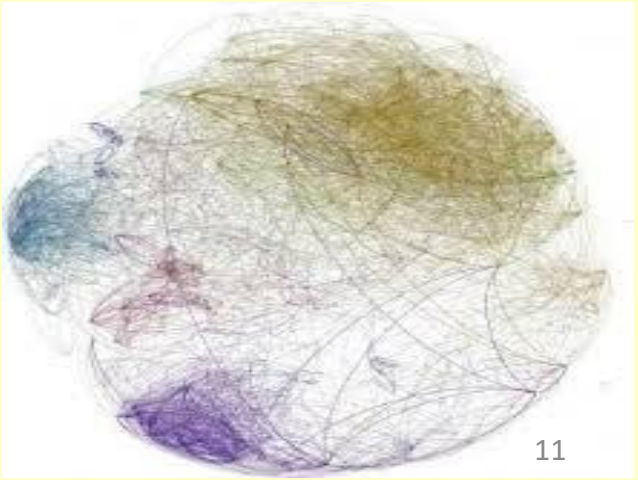
- Larger problems
- Poor project management
- Lack of adequate training in software engineering
- Increasing skill shortage and manpower turnover
- Low productivity improvements.

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In What Ways Management of Software Projects Differ from Management of Engineering Systems?



- **Software is Intangible:**
  - Software is invisible till it is ready and runs
- **Large Change Impact:**
  - A small change request can have terrible impact
  - Change is the rule rather than exception
- **Intellectual work…**



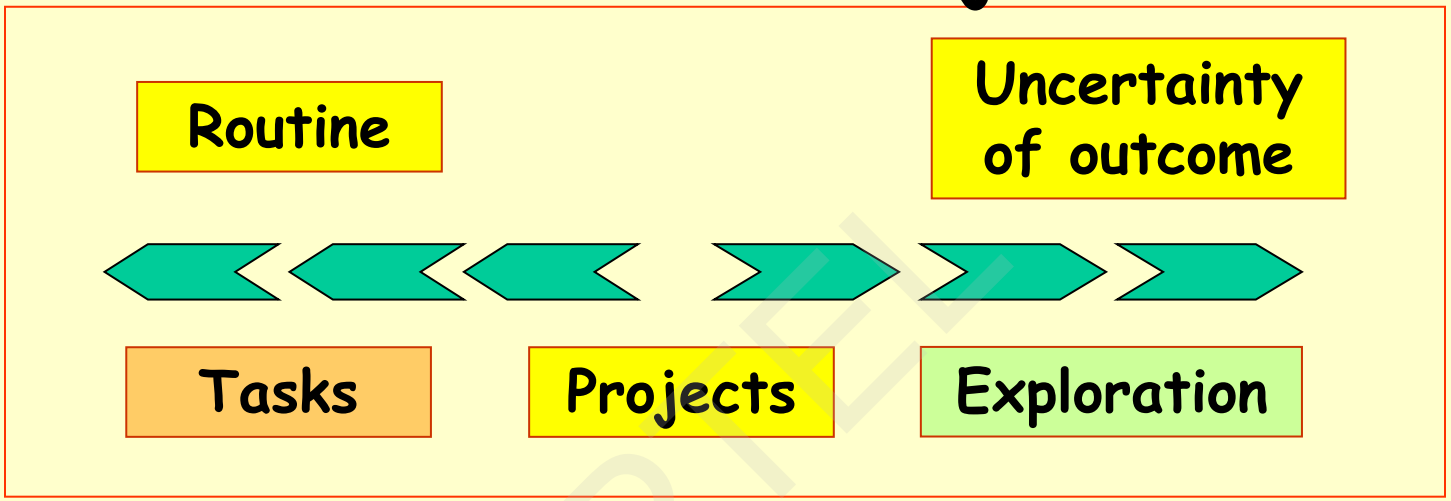
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Why Software Project Management is Hard?

- It is hard to manage anything that you cannot see…
- Management of intellectual and complex work…
- Changing customer requirements is the rule rather than exception…
- Manpower turnover…

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Jobs versus Projects



Tasks or Jobs – repetition of very well-defined and well understood tasks with very little uncertainty

Exploration – The outcome is very uncertain, e.g. finding a cure for cancer...

Projects – in the middle! Has challenge as well as routine ...
Therefore project Management Important...

What is a project?

- **Some dictionary definitions:**
  - ***"A planned undertaking"***
  - ***"A large undertaking, e.g. a public works scheme"***

- **Key points above are *planning* and *size* of task**

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More Precisely: What Is a Project?

- A project is "a temporary endeavor undertaken to create a unique product, service, or result" (PMBOK® Guide, Fifth Edition, 2012)
- A set of activities undertaken within a defined time period in order to meet a specific set of goals/objectives within a budget.
- A project generally exhibits most of the following conditions:
  - **It is of finite duration**
  - **Usually nontrivial**
  - **Non-repetitive**
  - **Requires resources**

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A Project…

- An endeavor with specific objectives:
  - Usually consists of multiple tasks
  - With defined precedence relationships
  - With a specific time period for completion
- **Non-Software Examples?**
  - A wedding
  - A BTech degree
  - A house construction project
  - A political election campaign

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What is a Task?

- **A small piece of work:**
  - Meant to accomplish a straightforward goal
  - Effort of no longer than a few person-hours
  - Involves only one or two people
  - May or may not be a part of some project
  - **Usually repetition of a previously accomplished task**
- **Non-software Examples:**
  - Attend a lecture class
  - Buy a chocolate from the market
  - Book a railway ticket

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Project Stakeholders

- **Project stakeholders are individuals and organizations that are actively involved in the project, or whose interests may be positively or negatively affected as a result of project execution or project completion; they may also exert influence over the project and its results.**
- **Key Stakeholders**
  - **Project manager** —the individual responsible for managing the project.
  - **Customer** —the individual or organization that will use the project's product or service.
  - **Project team members** —the group that is performing the work of the project.
  - **Sponsor**—the individual or group that provides the financial resources for the project.

Project Stakeholders

- **Internal project stakeholders:**
  - Project manager, Project team, top management
- **External project stakeholders:**
  - Project's customers (if they are external to the organization), vendors

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Project Scope

- It is what needs to be achieved and the work that must be done to deliver a project.
- A list of specific goals, deliverables, tasks, deadlines, and costs.

Project Success: Sounds of Failure's Foot Steps…

- Development team doesn't understand customer's needs.
- Project scope is poorly defined
- Changes poorly managed.
- Chosen technology changes.
- Business needs change.
- Unrealistic deadlines.
- Inexperienced team.
- Poor project management.

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W5HH Principle

- A series of questions that lead to a definition of project scope:
  - Why is the software being built?
  - What will be done?
  - When will it be done?
  - Who is responsible for a function?
  - Where are they organizationally located?
  - How will the job be done technically and managerially?
  - How much of each resource is needed?



Barry Boehm

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Types of Software Projects

- Two types of software projects:
  - Products (Generic software)
  - Services (custom software)
- Total business – appx. $3.5 Trillion
  - Half in products and half services
  - **Services segment is growing fast!**

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CONCEPTS COVERED

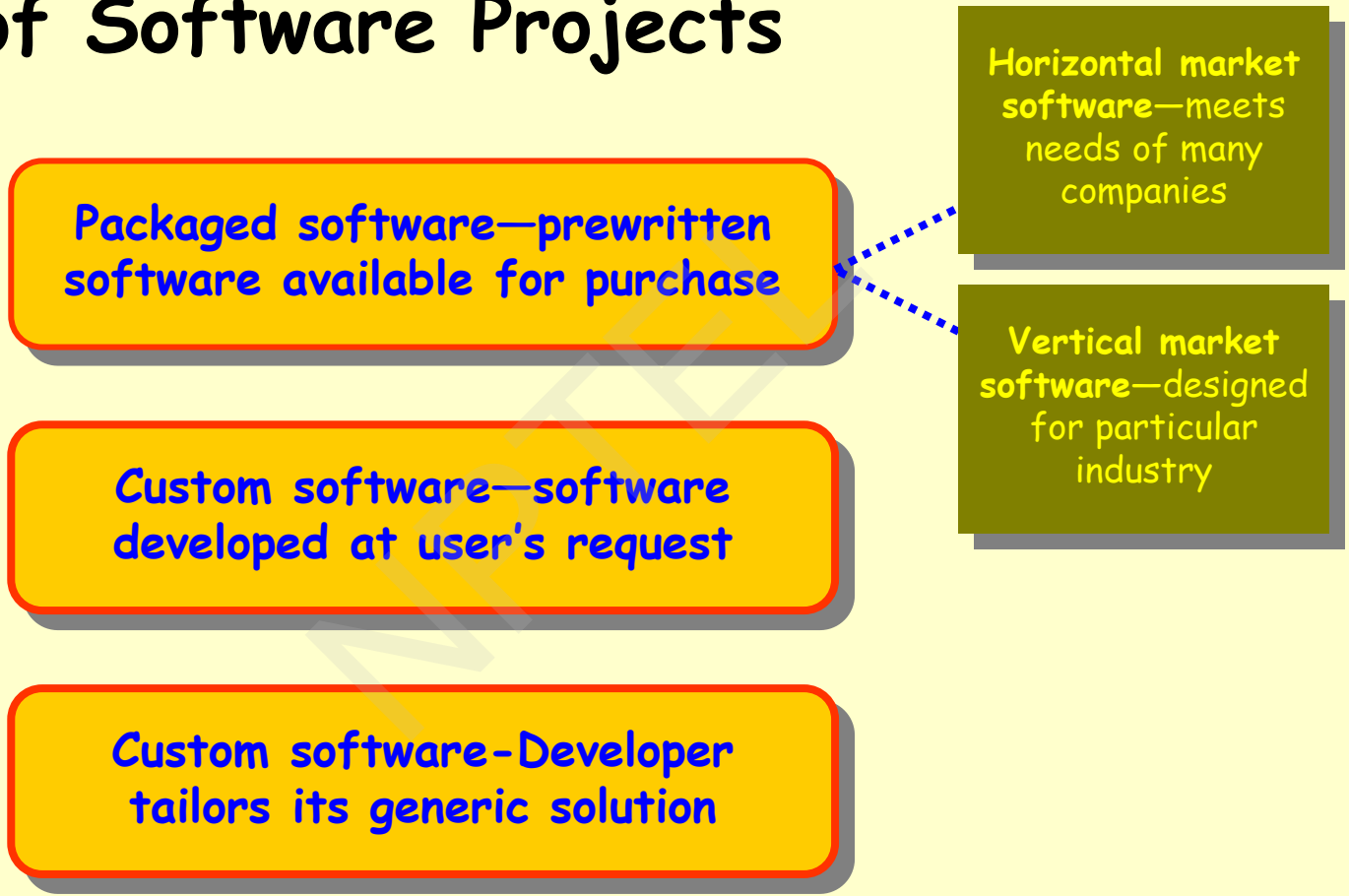
- Software products and services.
- Project management activities
- Traditional versus modern projects

Types of Software

- Software products
- Software services

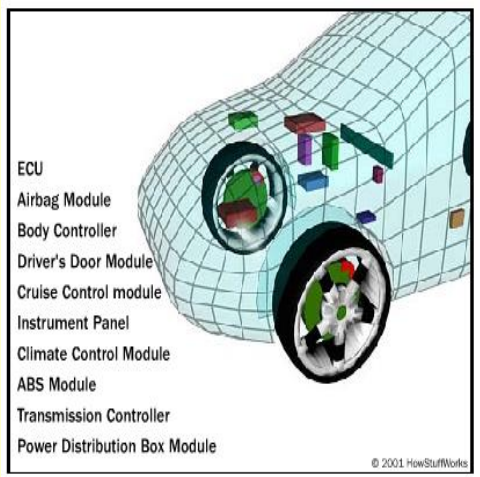
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Types of Software Projects



Another Software Classification

- **Information Systems:**
  - Store data, process data, present data and statistics
  - **Example:** MIS, Hotel Management software, Patient Management software, etc
  - **Web-based software versus Stand-alone software**



- **Embedded software:**
  - Controls hardware
  - **Example:** Automobile control software, Nuclear plant control software, robots, toys, TV remote…

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Software Services

- Software service is an umbrella term, includes:
  - **Software customization**
  - **Software maintenance**
  - **Software testing**
  - **Also contract programmers who carry out coding or any other assigned activities.**



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Factors responsible for accelerated gravitation towards services…

- **Lots of code is available in a company:**
  - New software can be developed by modifying the closest.
- **Business velocity has increased tremendously:**
  - Requires shortening of project duration
- **Program sizes have become large**

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Scenario of Indian Software Companies

Indian companies have largely focused on the services segment --- Why?

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Project Management Activities

Project Management = *Plan the work* and *work the plan*

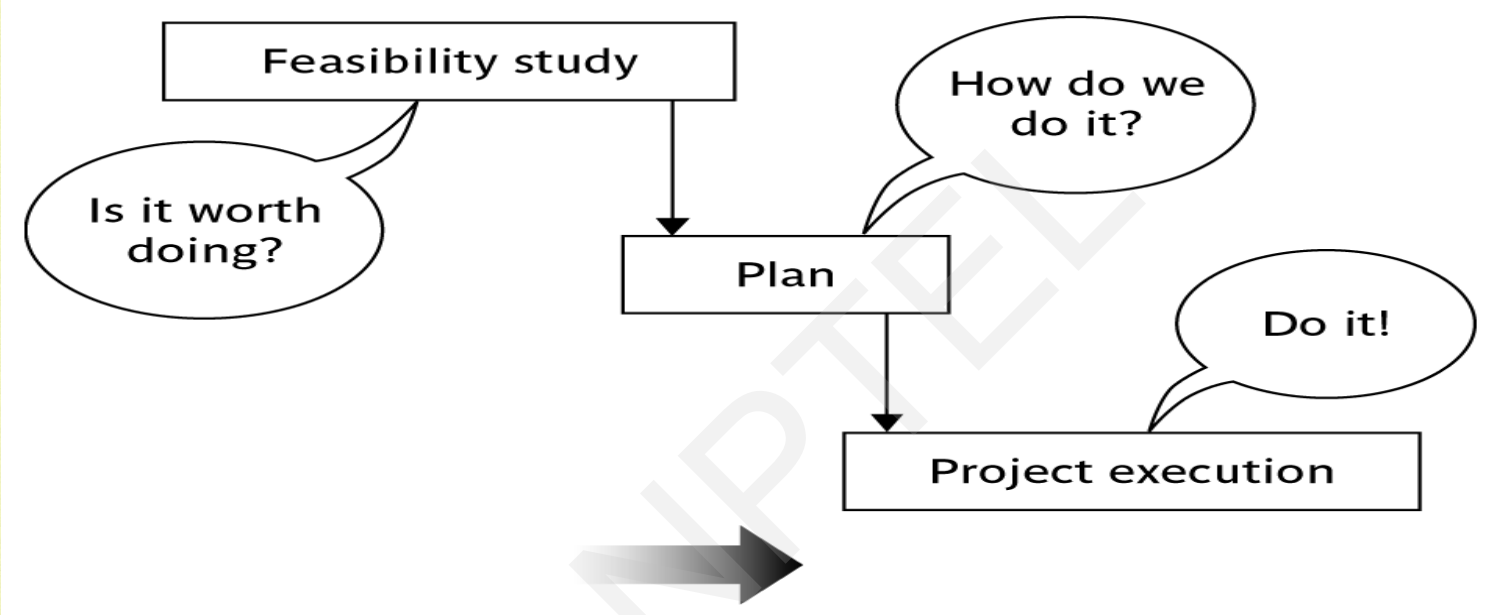
Planning: Estimation, resource allocation, and scheduling

Staffing: Recruiting and motivating personnel

Directing: Ensure team acts as a whole

Monitoring and Controlling: Detect plan deviations and take corrective actions

Major Responsibilities of a Project Manager



Major Responsibilities of a Project Manager

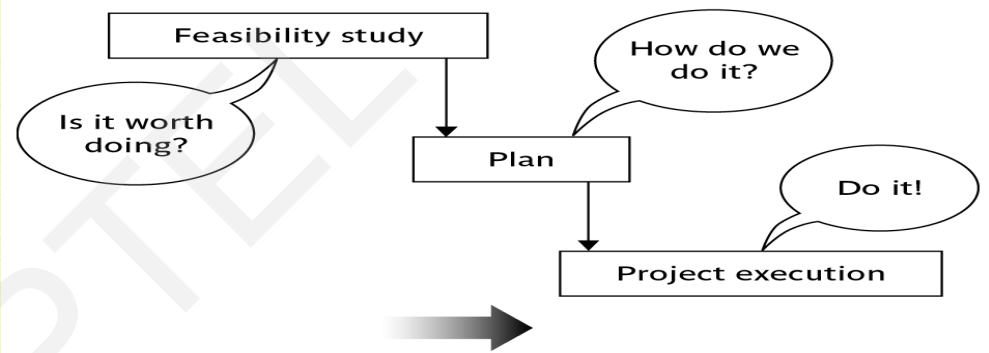
Feasibility study: Is the project technically feasible and worthwhile from a business point of view?

Planning: Only done if project is feasible

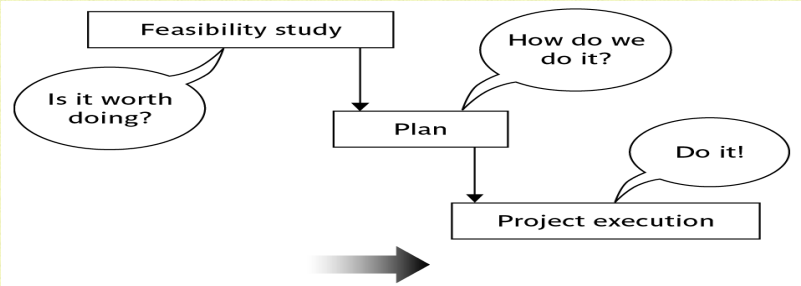
Execution monitoring and control: Monitor and control plan implementation, but plan may be changed as project proceeds….

Project Planning

- Carried out before development starts.
- Important activities:
  - **Estimation**
  - **Scheduling**
  - **Staffing**
  - **Risk management**
  - **Miscellaneous plans**

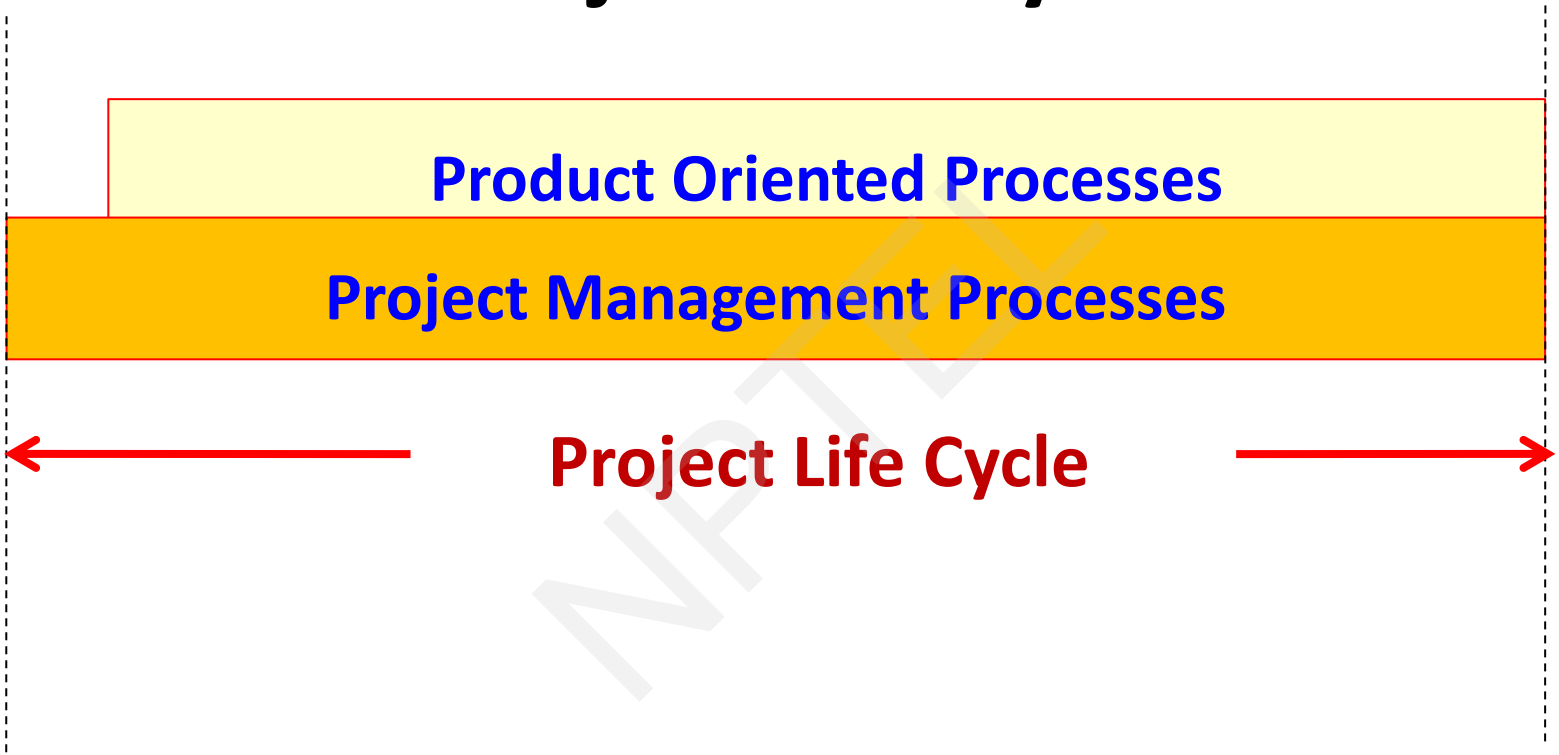


Monitoring and Control



- **Lasts for entire active project duration.**
- **Monitoring** – checking on progress, revising plans...
- **Controlling** – taking action to remedy hold-ups...
- **Innovating** – coming up with solutions when problems emerge...
- **Representing** – liaisoning with clients, users, developers and other stakeholders...

Project Life Cycle

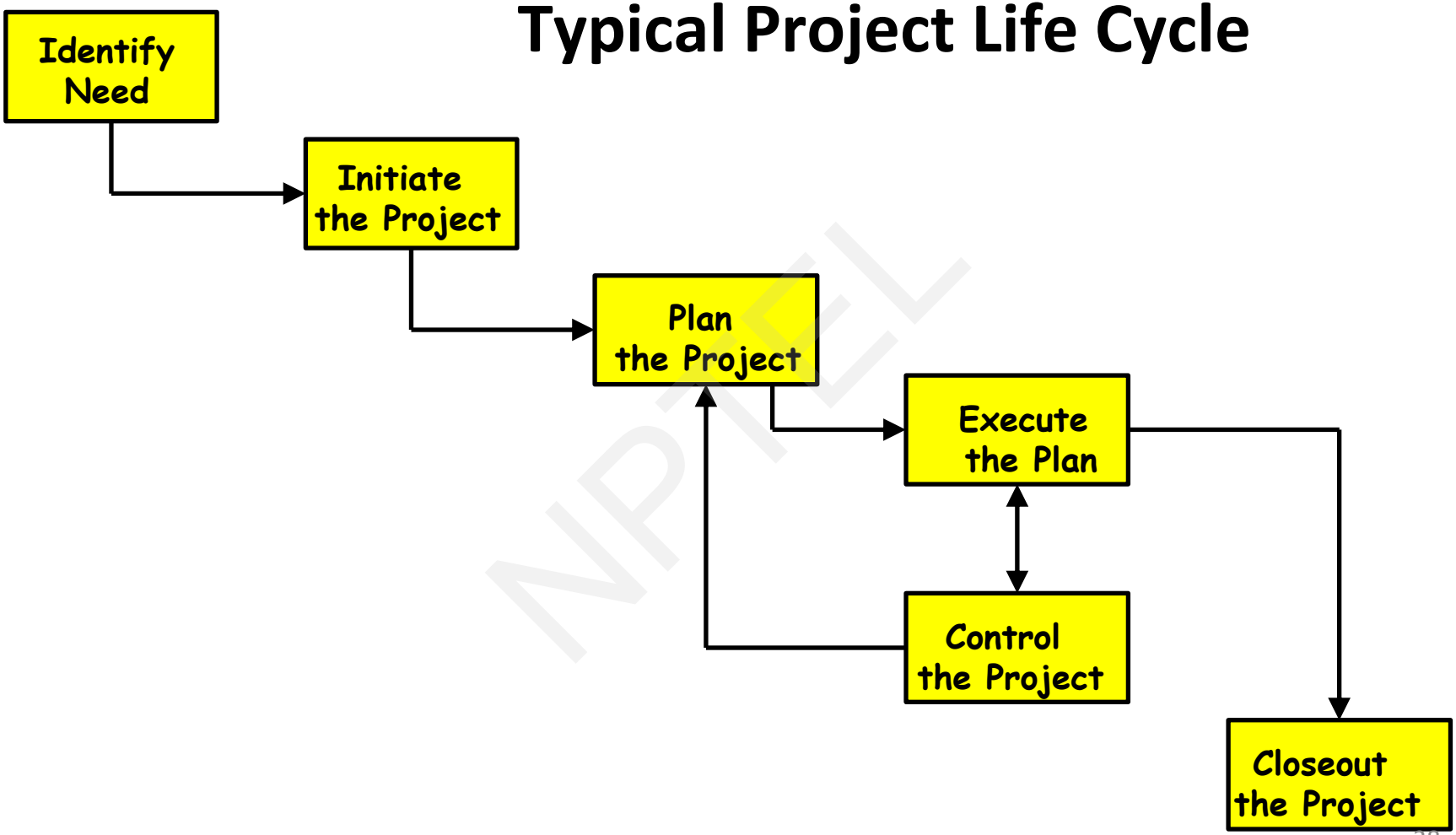


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- Project management standards
- Life cycle models

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Typical Project Life Cycle



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Product Development Processes versus Project Management Processes

Project Management Processes:

Concerned with planning and controlling the work of the project.

Product-oriented processes:

Concerned with specifying and creating the project products

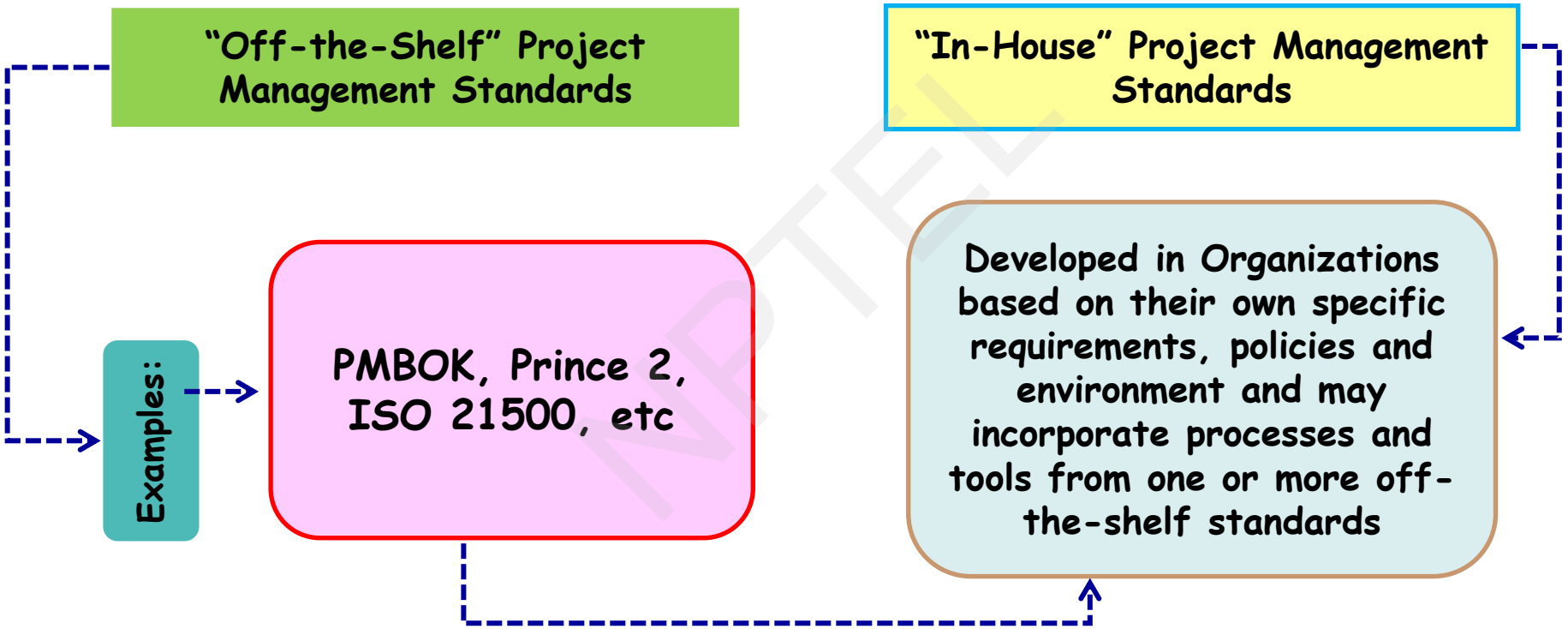
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Project Management Processes

- **Initiating processes**
  - Complete of the business case and project charter
- **Planning processes**
  - Complete the WBS and scope statement, project schedule and cost estimate
- **Executing processes**
  - Perform actions necessary to complete the work described in the plan
- **Monitoring and controlling processes**
  - Measure progress toward the project objectives, monitoring deviation from the plan and taking corrective action to match progress with the plan
- **Closing processes**
  - Formal acceptance of the work and creation of closing documents

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Major Project Management Standards
(Conventional Types and Customized)



What is PRINCE 2 ?

- PRINCE stands for: Projects IN Controlled Environments
  - A generic project management method--PRINCE1 was specific for IT-project management.
  - developed by the UK's Office of Government Commerce
- Popularly used in UK.
- Process-driven approach.
- PRINCE2 can be used for both small and big projects.

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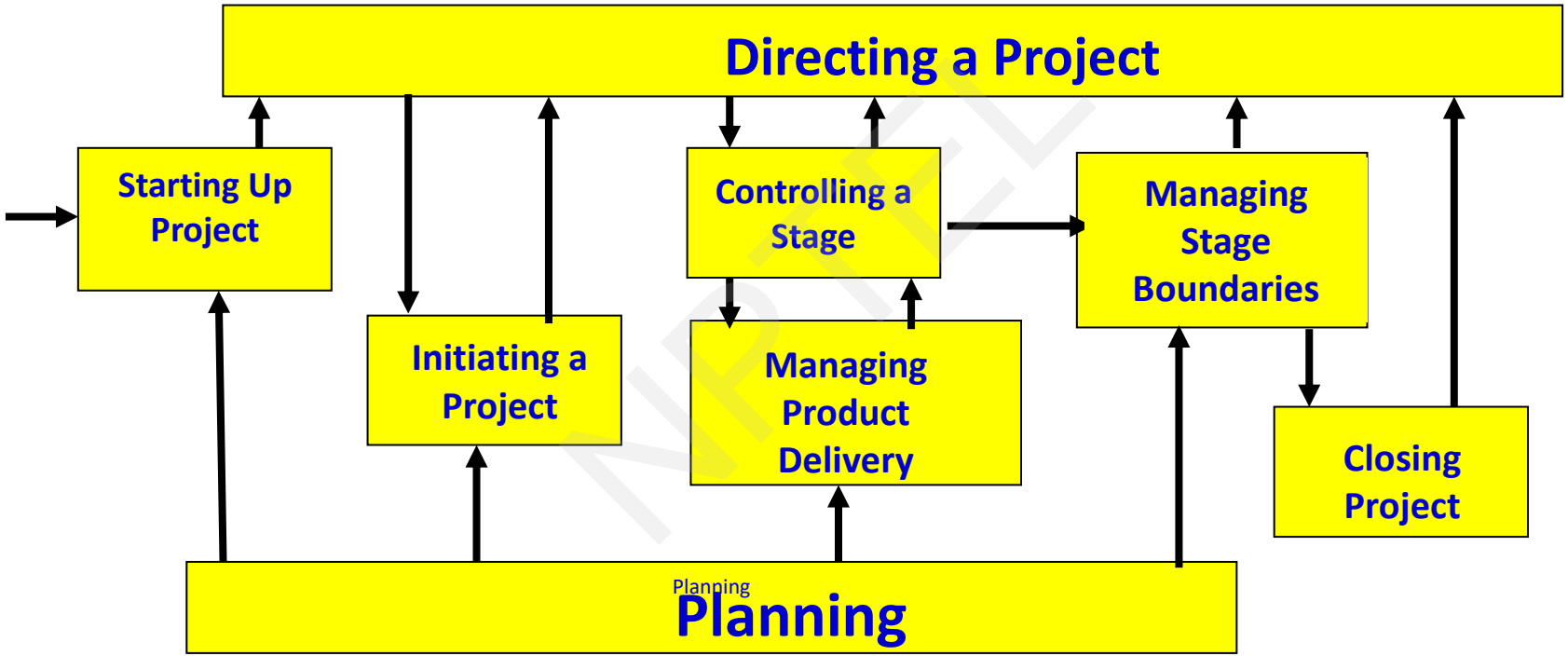
PRINCE2® – Processes

Seven processes.

- Starting a project
- Initiating a project
- Directing a project
- Managing a stage boundary
- Controlling a stage
- Managing product delivery

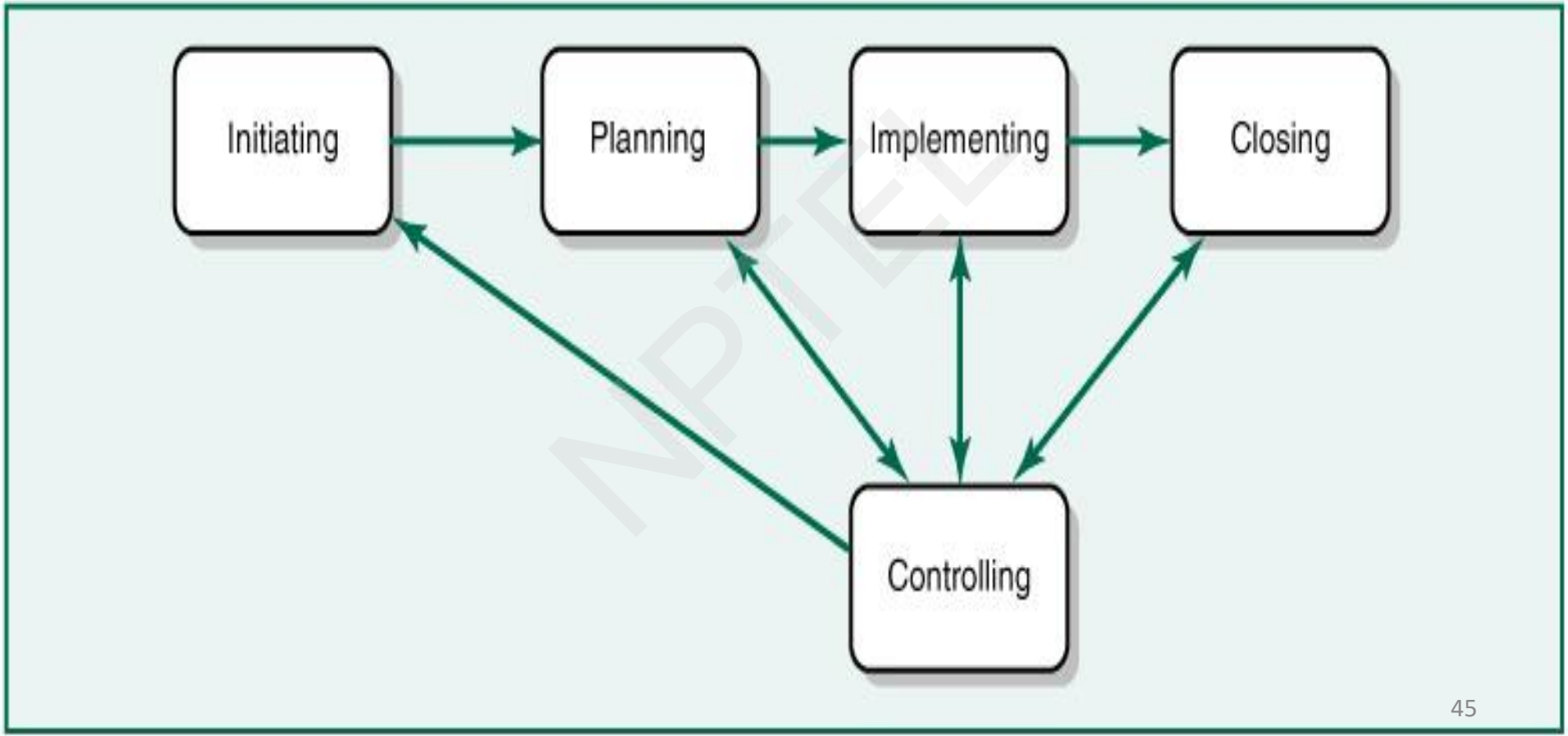
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PRINCE2 Processes



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ISO 21500 Life Cycle



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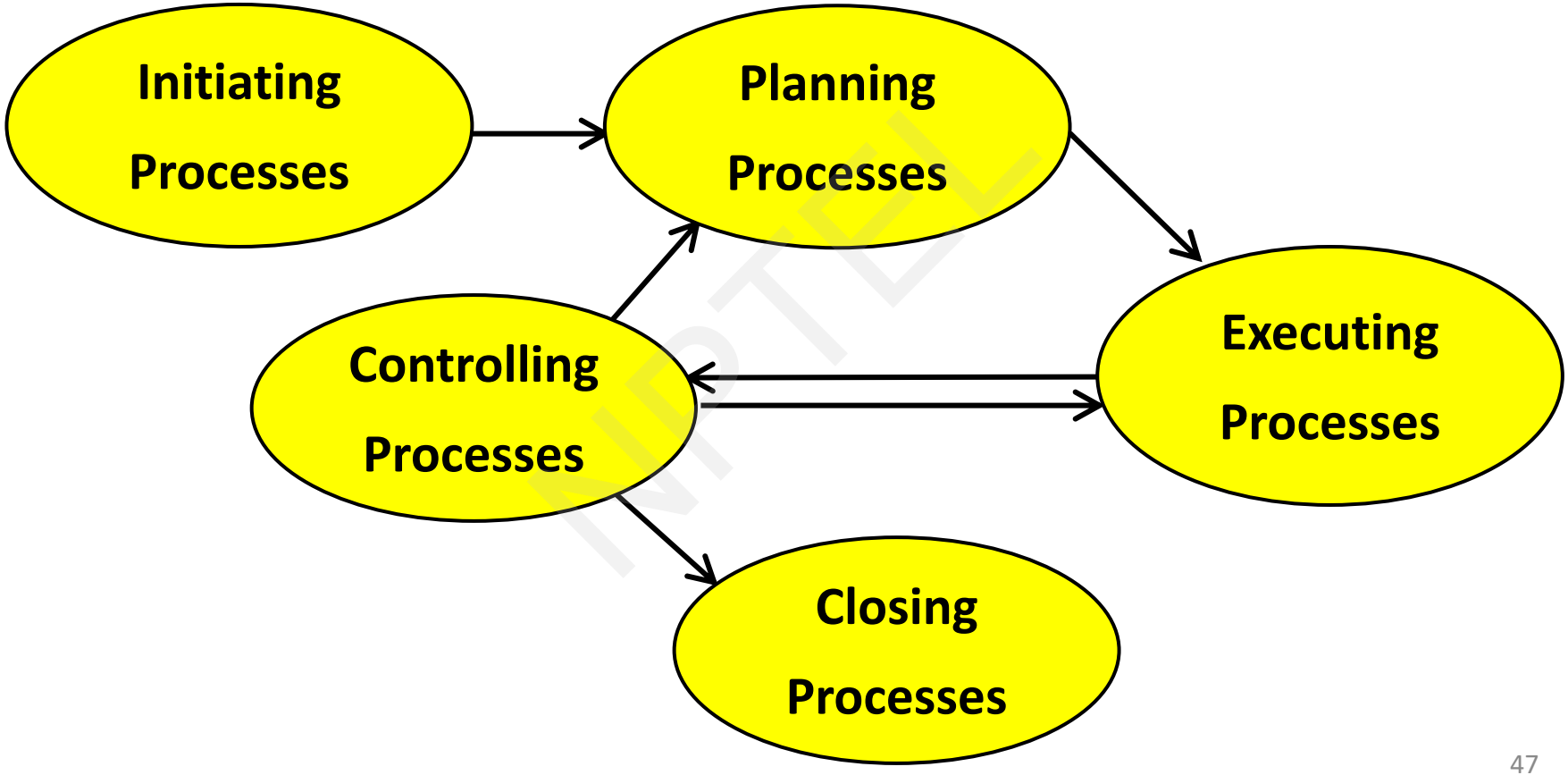
PMBOK

- Project Management Body of Knowledge:
  - The Dominant Project management standard
  - A set of standard terminology and guidelines (a body of knowledge) for project management
  - Documented in a book --- fifth Edition came out in 2013
  - Published by project Management Institute (PMI)
  - The *PMBOK Guide* is also used as a support to prepare the certifications offered by the PMI



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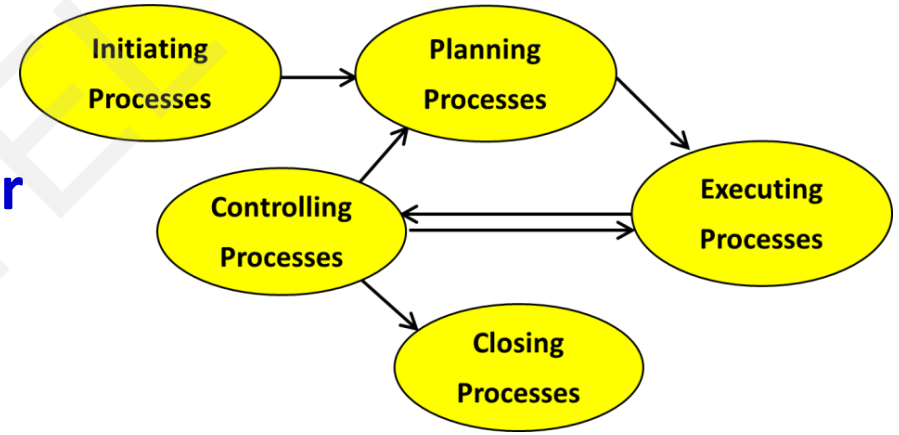
PMBOK Process Groups



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Project Initiation

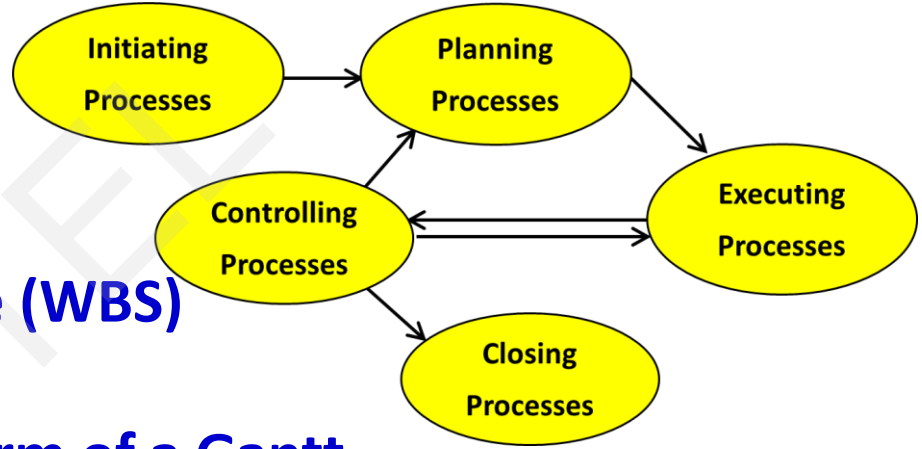
- The main goal is to formally select and start off projects
- Key outputs include:
  - **Assigning the project manager**
  - **Identifying key stakeholders**
  - **Completing the project charter**



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Project Planning

- The project plan *guides execution*
- Key outputs include:
  - A **work breakdown structure (WBS)**
  - **A project schedule, in the form of a Gantt chart with all dependencies and resources**
  - **A list of prioritized risks**



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PRINCE vs PMBOK

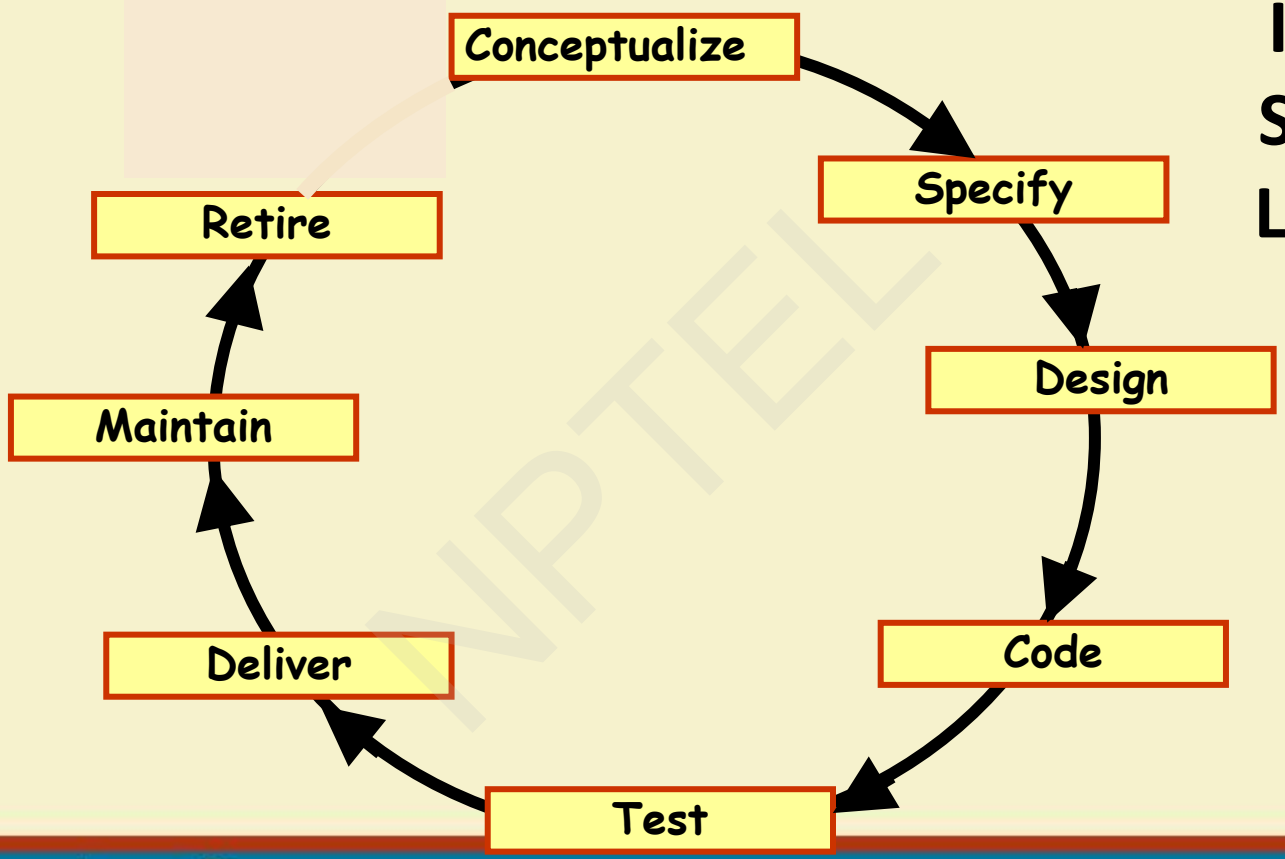
PRINCE2	PMBOK
Origin – United Kingdom	Origin – United States
Administered by APMG	Administered by PMI
Worldwide adoption	Worldwide adoption
A process based project management methodology	A knowledge based approach to project management
Prescriptive, but tailorable	Non-prescriptive
Used mostly within UK	Used worldwide and followed by over 75% of projects

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Life Cycle Models

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Intuitive Software Life Cycle



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Life Cycle Model

- A software life cycle model (also process model or SDLC):
  - –A descriptive and diagrammatic model of software life cycle:
  - –Identifies all the activities undertaken during product development,
  - –Establishes a precedence ordering among the different activities,
  - –Divides life cycle into phases.

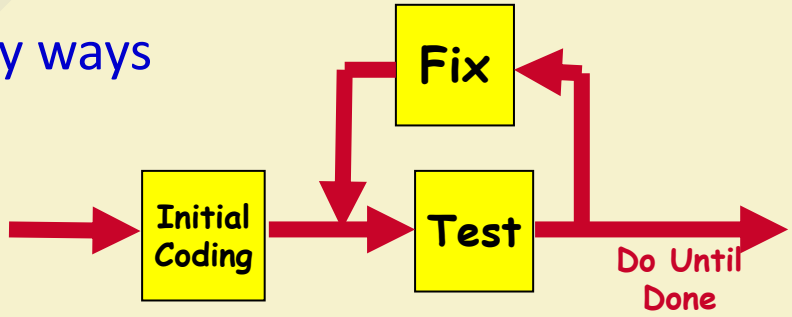
Life Cycle Model (CONT.)

- Each life cycle phase consists of several activities.
  - For example, the design stage might consist of:
    - structured analysis
    - structured design
    - Design review

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Build and Fix Model

- When a program is developed by a single programmer ---
  - The problem is within the grasp of an individual.
  - He has the freedom to decide his exact steps and still succeed --- called Exploratory model--- One can use it in many ways
  - Code ➔ Test ➔ Design
  - Code ➔ Design ➔ Test ➔ Change Code ➔
  - Specify ➔ code ➔ Design ➔ Test ➔ etc.



Life Cycle Model (CONT.)

- When software is being developed by a team:
  - There must be a precise understanding among team members as to when to do what,
  - Otherwise, it would lead to chaos and project failure.

Life Cycle Model (CONT.)

- A software project will never succeed if:
  - –one engineer starts writing code,
  - –another concentrates on writing the test document first,
  - –yet another engineer first defines the file structure
  - –another defines the I/O for his portion first.

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Phase Entry and Exit Criteria

- A life cycle model:



  - defines entry and exit criteria for every phase.
  - A phase is considered to be complete:
    - only when all its exit criteria are satisfied.

Phase Entry and Exit Criteria

- What is the phase exit criteria for the software requirements specification phase?
  - **Software Requirements Specification (SRS) document is complete, reviewed, and approved by the customer.**
- A phase can start:
  - Only if its phase-entry criteria have been satisfied.

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Life Cycle Model: Milestones

- Milestones help software project managers:
  - –Track the progress of the project.
  - –**Phase entry and exit are important milestones.**



Life Cycle and Project Management

- When a life cycle model is followed:
  - –The project manager can at any time fairly accurately tell,
    - At which stage (e.g., design, code, test, etc.) the project is.

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Project Management Without Life Cycle Model

- It becomes very difficult to track the progress of the project.
  - –The project manager would have to depend on the guesses of the team members.
- This usually leads to a problem:
  - –known as the **99% complete syndrome.**

Project Deliverables: Myth and Reality

Myth:

The only deliverable for a successful project is the working program.

Reality:

Documentation of all aspects of software development are needed to help in operation and maintenance.

Life Cycle Model (CONT.)

- Many life cycle models have been proposed.
- We confine our attention to only a few commonly used models.
  - Waterfall
  - V model,
  - Evolutionary,
  - Prototyping
  - Spiral model,
  - Agile models

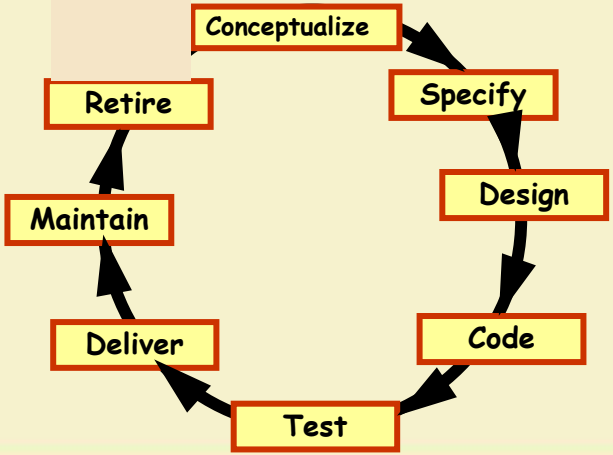
Traditional models

- Software life cycle (or software process):
  - Series of identifiable stages that a software product undergoes during its life time:
    - Feasibility study
    - Requirements analysis and specification,
    - Design,
    - Coding,
    - Testing
    - Maintenance.

Software Life Cycle

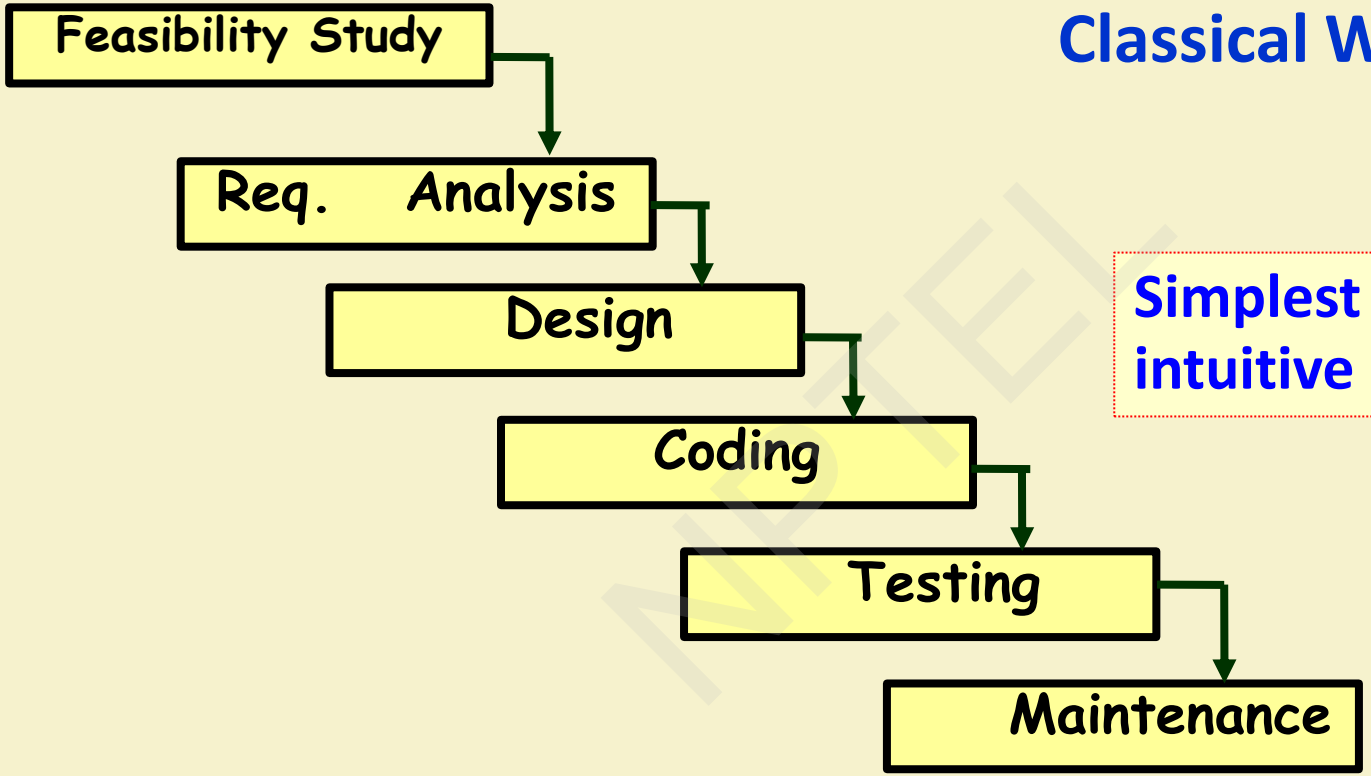
Classical Waterfall Model

- Classical waterfall model divides life cycle into following phases:
  - **Feasibility study,**
  - **Requirements analysis and specification,**
  - **Design,**
  - **Coding and unit testing,**
  - **Integration and system testing,**
  - **Maintenance.**



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Classical Waterfall Model

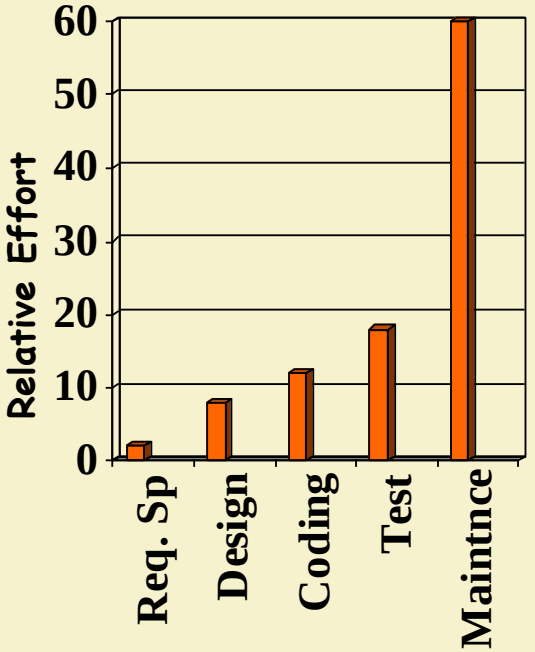


Simplest and most intuitive

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Relative Effort for Phases

- Phases between feasibility study and testing
  - Called development phases.
- Among all life cycle phases
  - Maintenance phase consumes maximum effort.
- Among development phases,
  - Testing phase consumes the maximum effort.



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- Most organizations usually define:
  - Standards on the outputs (deliverables) produced at the end of every phase
  - Entry and exit criteria for every phase.
- They also prescribe methodologies for:
  - **Specification,**
  - **Design,**
  - **Testing,**
  - **Project management, etc.**

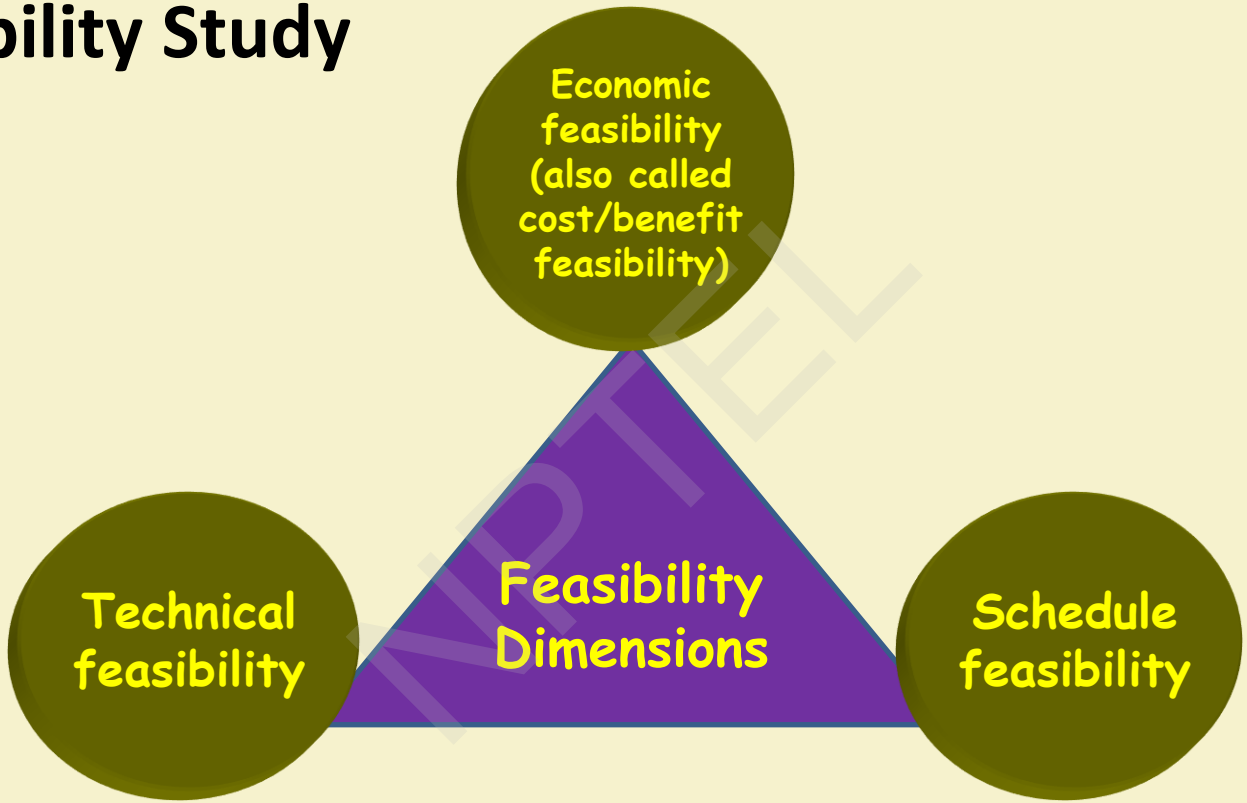
Process Model

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Classical Waterfall Model (CONT.)

- The guidelines and methodologies of an organization:
  - –Called the organization's software development methodology.
- Software development organizations:
  - – Expect fresh engineers to master the organization's software development methodology.

Feasibility Study



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- Main aim of feasibility study: determine whether developing the software is:
  - **Financially worthwhile**
  - **Technically feasible.**

Feasibility Study

- Roughly understand what customer wants:
  - **Data which would be input to the system,**
  - **Processing needed on these data,**
  - **Output data to be produced by the system,**
  - **Various constraints on the behavior of the system.**

First Step

- SPF Scheme for CFL
- CFL has a large number of employees, exceeding 50,000.
- Majority of these are casual labourers
- Mining being a risky profession:
  - Casualties are high
- Though there is a PF:
  - But settlement time is high
- There is a need of SPF:
  - For faster disbursement of benefits

Case Study

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Feasibility: Case Study

- Manager visits main office, finds out the main functionalities required
- Visits mine site, finds out the data to be input
- Suggests alternate solutions
- Determines the best solution
- Presents to the CFL Officials
- Go/No-Go Decision

Activities During Feasibility Study

- Work out an overall understanding of the problem.
- Formulate different solution strategies.
- Examine alternate solution strategies in terms of:
  - resources required,
  - cost of development, and
  - development time.

- Perform a cost/benefit analysis:
  - Determine which solution is the best.
  - May also find that none of the solutions is feasible due to:
    - high cost,
    - resource constraints,
    - technical reasons.

Activities during Feasibility Study

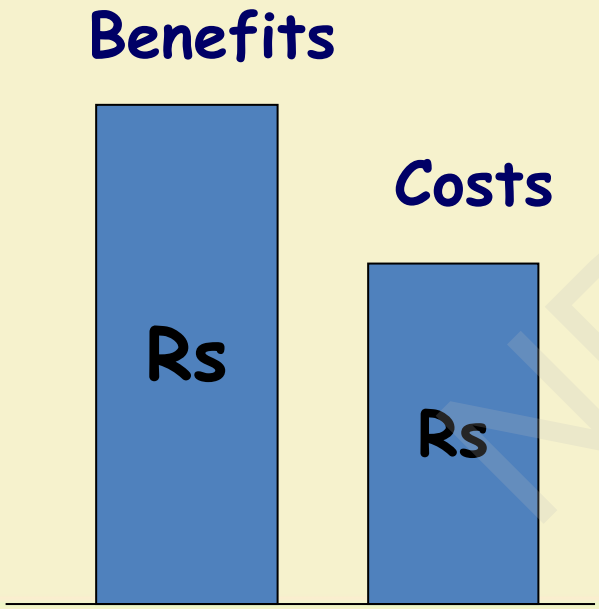
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Cost benefit analysis (CBA)

- Need to identify all costs --- these could be:
  - **Development costs**
  - **Set-up**
  - **Operational costs**
- Identify the value of benefits
- Check benefits are greater than costs

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The business case



- Benefits of delivered project must outweigh costs
- Costs include:
  - Development
  - Operation
- Benefits:
  - Quantifiable
  - Non-quantifiable

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The business case

- Feasibility studies should help write a ‘business case’
- Should provide a justification for starting the project
- Should show that the benefits of the project will exceed:
  - **Various costs**
- Needs to take account of business risks

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Writing an Effective Business Case

1. **Executive summary**
2. **Project background:**
   - The focus must be on what, exactly, the project is undertaking, and should not be confused with what might be a bigger picture.
3. **Business opportunity**
   - What difference will it make?
   - What if we don't do it?
4. **Costs**
   - Should include the cost of development, implementation, training, change management, and operations.
5. **Benefits**
   - Benefits usually presented in terms of revenue generation and cost reductions.
6. **Risks**
   - Identify risks.
   - Explain how these will be managed.

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Iterative Waterfall Model

- Classical waterfall model is idealistic:
  - –Assumes that no defect is introduced during any development activity.
  - –In practice:
    - Defects do get introduced in almost every phase of the life cycle.

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Iterative Waterfall Model (CONT.)

- Defects usually get detected much later in the life cycle:
  - For example, a design defect might go unnoticed till the coding or testing phase.
  - **The later the phase in which the defect gets detected, the more expensive is its removal --- why?**

Iterative Waterfall Model (CONT.)

- Once a defect is detected:
  - The phase in which it occurred needs to be reworked.
  - Redo some of the work done during that and all subsequent phases.
- Therefore need feedback paths in the classical waterfall model.


Thank you